AN ECLECTIC TECHNOLOGIES FOUNDATIONAL DEFINITION

FD-DBL — Design Basis Layers

Foundational Definition: The Three Layers of Design Basis Analysis, the Function That Holds Them Together, the Assumptions Each Layer Carries, the Adversary Characteristics That Relax Them, the Change in What Success Means, and the Operating Limits That Keep Operation Inside the Design Basis

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Companion root definitions: FD-BL (the Bright Line); FD-LD (layer decomposition and determinism); FD-EV (the envelope); FD-BR (the Broker); FD-LS (local sovereignty). FD-DBL owns the design basis layers; it owns none of the companions' vocabulary.
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1. Purpose and the Defect This Instrument Corrects

Every series in the corpus is built on the same analytical progression, borrowed from nuclear safety. First comes the Design Basis, which establishes the function a facility performs. Then comes the Design Basis

Accident, which adds the disruptive events the facility must be designed to withstand. Then comes the military action extension, which adds an adversary. The DBA-MA series is named for the third layer; the DBA-EN series describes itself as applying “the same three-layer analytical progression that nuclear safety has used since the 1960s”; and the sector frameworks inherit the progression from both.

The progression is used everywhere and defined nowhere. Four defects follow.

First, the characteristics that distinguish the adversary layer are enumerated differently in each instrument that states them. The MA Introduction (v1.9) names four at its §3: intentionality, persistence, precision, and denial. The MA International Framework (v2.0) names four different ones at its §2.3 — simultaneous occurrence, deliberate causation, indefinite duration, and systematic degradation of external support — and three at its §8.2. The MA Accident Description (v3.0) names four at its §1.3, with adaptive threat behavior standing apart from intentional causation, and three at its §2.1. None of these contradicts the others in spirit, but a defining set that changes size and membership between a parent and its children is the drift the Foundational Definitions family exists to prevent. It also blocks a necessary judgment: until the set is fixed, no one can say which results established at the accident layer survive into the adversary layer.

Second, the most consequential feature of the progression has never been stated as a definition. At the first two layers, success means the function is delivered. At the adversary layer, success means something else: when the function cannot be delivered, the facility fails in a manner that keeps consequences bounded. The corpus enacts this change — the Autonomous Safe Shutdown Condition defined in the MA Accident Description is its nuclear instance — but nowhere says that the meaning of “the design works” changes between layers.

Third, the relationship between layers is carried by a single word or a single arrow (“builds on”; DB → DBA → DBA-MA) that does three different jobs. Sometimes it means that one layer specializes another, sometimes that one inherits another’s method, and sometimes that one was historically derived from another. These are different relations with different consequences, and a reader cannot tell which is meant.

Fourth, a design basis is not only analyzed; it is operated. The people, processes, and technologies that run a facility must keep it inside its design basis, and they do so against a ladder of limits, each paired with a response and, at some rungs, a report to a regulator. The corpus has never stated that ladder as part of the design basis it inherits. Without it, the layers say what the design must withstand but not how operation is kept within the design, or how anyone would know that it had left.

This instrument defines what an analytical layer is, defines the three layers, states the assumptions the accident layer carries, fixes the four adversary characteristics as the canonical set, defines the change in success criterion, states the relations between the layers, defines the operating limits that travel with the design basis, and states how cyber runs through every layer.

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2. Authority, Family, and Convention

2.1 Precedence

This instrument owns the definition of the design basis layers for all documents in the corpus. Foundational Definitions take precedence over frameworks, sector documents, facility-class

instruments, and implementation guidance. A framework may instantiate a layer for its sector — naming the function, the event set, the threat assessment, and the protected state — but it shall not vary the layer definition (§4.1), the stated assumptions (§4.3), the adversary characteristics (§4.4), the success criteria (§4.5), or the relations between layers (§4.6). Propagation shall be by reference to this instrument, not by restating text.

2.2 The Foundational Definitions family

The Foundational Definition: Design Basis Layers (FD-DBL) qualifies under the family membership test: the correctness of every DBA-MA instrument, and of every sector instrument that carries a military action extension, depends on agreeing what kind of thing a design basis layer is, which assumptions the adversary removes, and what success means at each layer. It is a root definition. It has dependents and no parent.

2.3 Declaration convention

Per the family convention, this instrument states what kind of thing each of its constructs is before defining it. The declarations are collected here so that the whole of the instrument's ontology can be read in one place; each is defined in the section named.

A design basis layer is an analytical construct with five parts (§4.1). It is not a document, not a series, and not a regulatory category. It is not an architectural layer in the sense of FD-LD (*Layer Decomposition and Determinism*), where "two-layer architecture" is reserved for the infrastructure and cognitive decomposition, and it is not a boundary between architectural layers, which FD-BL (*The Bright Line*) rules out for the Bright Line. It is not a level of defense in depth in the IAEA sense; the levels of defense in depth are barriers and provisions within a design, while a design basis layer is a set of conditions under which a design is analyzed.

The function is the service function, the thing the facility is built to do (§4.2). It is the invariant of the progression.

Safety is a requirement on how the function is performed, not the function itself (§4.2). It is the one obligation that survives every layer unchanged (§4.5).

An assumption of the accident layer (A1, A2, or A3) is a stated premise under which the accident layer's events are postulated (§4.3). It is not a design requirement.

The adversary is an actor defined by capability, not by identity (§4.4).

A characteristic of the adversary — intentionality, persistence, precision, or denial — is a way of acting on the accident layer's event set, not a further event added to it, and each removes one of the stated assumptions (§4.4).

A success criterion is the test against which a design is judged at a given layer (§4.5).

A protected state is the condition a facility reaches and holds when its function cannot be delivered, in which consequences remain bounded (§4.5). Each sector, and sometimes facility classes within a sector, names its own.

The operating limits are a ladder of nested limits, each paired with who acts, what the action is, and what is reported (§4.7).

Cyber is a medium that runs through every layer, as people, process, and technology do (§4.9). It is not a layer.

Where this instrument uses the short word “layer,” it means a design basis layer and nothing else.

Citation form. Where the running text names another instrument, it gives the designation with the title in parentheses, as in FD-BR (*The Broker*), so the reader holds both. Where it cites a section, it gives the title and the section alone, as in *The Broker* §4.2 or the MA Accident Description at §1.3, so the section reads as part of a named document. The DBA-MA series is cited by short title: the MA Introduction, the MA International Framework (L1), the MA Accident Description (L2), the MA Trigger Framework (L3), and the MA Monitoring Guidance (L4). Full titles are in §9.

2.4 What this instrument does not alter

Nothing here alters the six threat categories of the MA International Framework (§2.5), the seven engineering design principles (MA International Framework §5.1–§5.7; MA Accident Description §4.1), the definition and acceptance criteria of the Autonomous Safe Shutdown Condition (MA Accident Description §3), the facility-class design-basis durations, the MA Trigger Framework, or the MA Monitoring Guidance. It does not decide where military action sits in the IAEA standards hierarchy; the MA International Framework at §8.1 places it as a category of design extension condition, and that placement stands. It does not set the parameters of any adversary envelope, which belong to each facility’s threat assessment. It does not alter any facility’s technical specifications, cybersecurity program, or any regulator’s reporting requirements; §4.7 describes how those instruments relate to the layers and cites the nuclear regime as its instance. This instrument is definitional. It states what the layers are and how they relate, not the engineering content of any layer.

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3. The Governing Questions

Invariant. What holds constant from one layer to the next?

Assumptions. What does the accident layer assume about the events it postulates?

Adversary. What distinguishes the adversary layer, and which assumptions does it remove?

Success. What does it mean for a design to succeed at each layer?

Relation. How is each layer related to the one before it?

Limits. How is operation kept inside the design basis, and what happens, and who is told, as it approaches or crosses a limit?

The adversary and success questions are the pair most often collapsed. A corpus that treats the adversary layer as “the accident layer with worse events” will look for a harder version of the same success criterion, and will not find the criterion that actually governs.

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4. The Definition

4.1 What kind of thing a layer is

A design basis layer is an analytical construct with five parts: the function the facility must perform; the set of conditions under which the design is analyzed; the assumptions under which those conditions are postulated; the success criterion against which the design is judged; and the operating limits, with their responses and reports, by which operation is kept inside the design (§4.7).

Each layer after the first keeps the function of the layer before it and changes one or more of the other parts. That is what makes the progression a progression rather than a collection of separate analyses.

4.2 The Design Basis layer

The Design Basis layer establishes the function. The function is the service function: the thing the facility is built to do. A power plant is built to generate power, a water system to deliver water, a pipeline to move gas. A facility is designed to perform a function that is critical by some stated measure, and it performs that function through people, process, and technology, with measures of how well the function is performed. The Design Basis names the function, the people, process, and technology that deliver it, and the measures of its performance. It also establishes the limits within which the function may be performed, from which the operating limits of §4.7 are derived.

Safety is not the function. It is a requirement on how the function is performed. Nothing is built simply to be safe; it is built to do something, and it must do that thing safely, to protect life, the investment, the earning potential of the asset, or for any number of other reasons the owner and the regulator recognize. The safety functions of a facility exist to meet that requirement, and they are means, not ends.

The function is the invariant of the progression. No later layer alters it. Later layers change only the conditions under which the function must be delivered, the assumptions under which those conditions are postulated, and what counts as success when the function cannot be delivered. The Design Basis is therefore the only layer that defines; every later layer constrains.

4.3 The Design Basis Accident layer

The Design Basis Accident layer adds a set of disruptive events: natural hazards such as fire, flood, earthquake, and storm; the failure of technology; the error of an operator; and the flaw in a process that was not anticipated. The set is drawn from operating experience — from events that have happened to someone's facility — and it is bounded because experience is bounded. A facility is designed so that, when any event in the set occurs, it can either keep performing its function or reach a safe, known condition.

The accident layer postulates its events under three assumptions. They are stated here because they are what the adversary layer removes.

A1, independence. Initiating events are independent of one another and indifferent to the facility's response, except for common-cause failures, which the accident layer identifies, analyzes as such, and addresses through diversity and defense in depth. An earthquake does not change its behavior because operators are restoring cooling, and equipment failures do not coordinate to defeat several safety systems at once; where a single cause can defeat several, the analysis says so and the design answers it.

A2, bounded progression. Each event follows a defined progression of initiation, response, and recovery, and ends.

A3, available support. External support — off-site power, emergency response, supply chains, staffing pipelines, communications, and regulatory oversight — remains available to the response.

The success criterion at the accident layer has two outcomes. The response either recovers operation to within the operating limits, so that the function continues, or brings the facility to a safe, known condition. The safe condition at this layer is a waypoint. It is reached with external support available (A3), and it is the starting point for recovery and eventual return to service.

4.4 The adversary layer

The adversary layer adds an adversary: an actor who sets out to defeat the facility's function, or a requirement on how the function is performed, of which safety is the one every facility carries (§4.2). The adversary may stop the function, seize the facility and take its function from those it is performed for, or turn the facility's plant or material against others. Each defeats the function or a requirement on it, and none redefines the function (§6.4). The adversary is defined by capability, not by identity. Any actor that exhibits all four of the characteristics below, at the level characterized in the facility's threat assessment, is within this layer. A nation-state military force, a state-equivalent force, and a well-resourced non-state group may each qualify. Identity informs the assessment of capability; it does not decide membership.

The four characteristics are the canonical set. They are not further events to be added to the accident set. Each is a way of acting on the accident set, and each removes one of the accident layer's assumptions.

Intentionality. The adversary chooses which conditions occur, when, and in what order, observes the facility's response, and adapts to defeat it. Adaptive behavior is part of intentionality, not a separate characteristic. Intentionality removes A1 in its first sense: the conditions are no longer indifferent to the response.

Persistence. The adversary sustains or repeats conditions beyond any defined progression. The event becomes an environment, in which the facility must keep operating safely for as long as the condition lasts. Persistence removes A2.

Precision. The adversary directs conditions at the specific elements whose loss produces the most difficult recovery path — the elements a careful reader of the facility's safety analysis would choose. Failures become correlated by design. Precision removes A1 in its second sense: failures no longer occur independently of one another, and the common-cause failures the accident layer analyzed as chance events become targets. From the adversary's side, a common cause is horizontally scalable: one action that defeats one instance of an identical component defeats every instance that shares it, across redundant trains, across units, and across facilities.

Denial. The adversary removes the mitigation and recovery that the response depends on: access to the site, repair, resupply, external emergency response, and communications. Denial removes A3.

Simultaneous challenge to several safety functions, which the MA International Framework lists at §2.3, is the result of intentionality and precision acting together. It is not a fifth characteristic.

4.5 The success criterion

The success criterion changes at the adversary layer.

At the Design Basis layer, success is delivery of the function within the operating limits.

At the accident layer, success is either recovery to within the operating limits or arrival at a safe, known condition from which recovery proceeds with external support (§4.3).

At the adversary layer, recovery is no longer a success outcome, because the adversary can defeat it and persistence and denial remove the support it relies on. Success is the transition to, and the holding of, a protected state in which consequences remain bounded, using only resources present at the facility, for as long as the adversary's action lasts.

The military-action tier marks a division of responsibility as well as a layer. Where the law of the asset's jurisdiction places defense against a state's armed forces with the state, as 10 CFR 50.13 does for nuclear facilities in the United States, reaching the tier moves to the state the duty to defend the facility against its attacker: to deter the attack, to intercept it and to end it. It does not move the duty this criterion states. The facility still reaches and holds its protected state using only resources present at the facility, because the state's defense is itself external support, and external support is the assumption (A3) that denial removes. An adversary able to deny a facility its off-site power, resupply and communications may deny the state's response as well, and that response may also be degraded by the attack, divided by a boundary of authority, or limited by policy, priority or resources, none of which the facility controls. The handoff of the duty to defend is therefore an allocation of responsibility and never a credit in the design. No design takes credit for a response by the state that it cannot show will arrive.

The change is therefore narrower than a replacement of one criterion by another. The accident layer already admits a safe condition as success. What changes is how that condition is held. At the accident layer it is a waypoint, reached with support and left by recovery. At the adversary layer it is self-sustaining and indefinite, and no path back is assumed. A design whose safe condition depends on off-site power, resupply, or outside repair meets the accident-layer criterion and fails the adversary-layer one.

Which functions can hold a protected state once denial has removed external support and authority is a separate question. FD-LS (*Local Sovereignty*) defines one class of such functions, the protective functions that need no authority from any reachable system to act; this instrument does not decide which functions must have that property.

The distinction between the function and the safety requirement is what makes this criterion coherent. At the adversary layer the facility may give up its function. It never gives up the requirement that whatever it does, including ceasing to perform its function, is done safely. The safety requirement is the one obligation that survives every layer unchanged.

Each sector names its protected state. The nuclear instance is the Autonomous Safe Shutdown Condition (MA Accident Description §3), in which the facility's service function, the generation of power, is given up and its safety functions are maintained by local means for as long as the military action lasts.

4.6 Relations between the layers

The arrow written DB → DBA → DBA-MA carries four distinct relations. Each instrument that uses the arrow shall be read as asserting all four, and none beyond them.

The function is preserved. Each layer keeps the function of the layer before it (§4.2).

Conditions widen; admissible designs narrow. Each layer analyzes the design under more conditions than the layer before it, so fewer designs satisfy it. The two move in opposite directions. When a later

layer is said to “build on” an earlier one, it means the later layer’s conditions are wider and the set of designs that satisfy it is narrower.

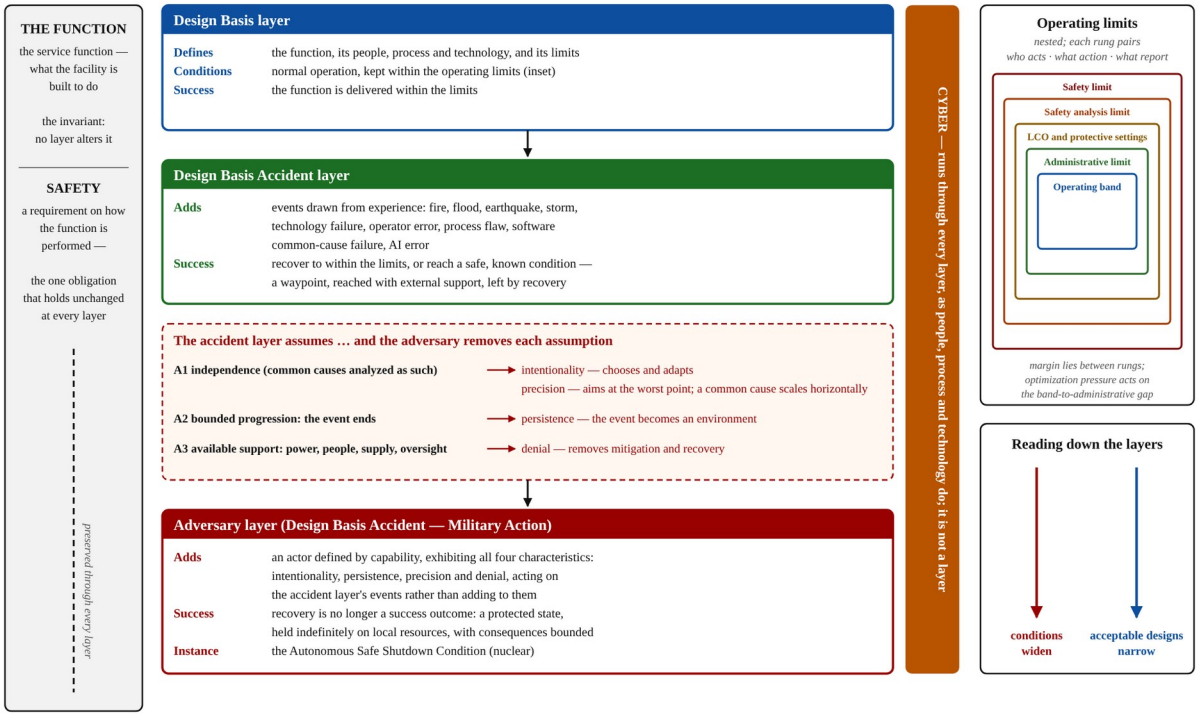
Method is inherited; category is not. The adversary layer uses the analytical method of the accident layer: defined conditions to be withstood, credited structures, systems, and components, and explicit acceptance criteria. The name “Design Basis Accident — Military Action” records that inheritance. It does not place the adversary layer within the regulatory category of design basis accidents; that placement is decided elsewhere (MA International Framework §8.1).

Results carry forward only where the assumptions are not relied on. A result established at the accident layer carries into the adversary layer unchanged if it does not depend on A1, A2, or A3. A result that depends on any of them must be re-established at the adversary layer. Emergency operating procedures written on the assumption that site access and external support are available (A3) are the standing example.

Figure 1 draws the layers and their relations on one page: the function and the safety requirement preserved through every layer; the three layers with their conditions and success criteria; the accident layer’s assumptions and the characteristics that remove them; cyber running through all three; the operating limits; and the widening of conditions against the narrowing of acceptable designs. The figure illustrates the sections it names and does not add to them; where the figure and the text differ, the text governs.

Figure 1 — The design basis layers at a glance

FD-DBL (Design Basis Layers): one function, three layers, the assumptions the adversary removes, cyber across all, and the limits operation is kept within



The arrow DB → DBA → DBA-MA carries four relations: the function is preserved; conditions widen while acceptable designs narrow; the adversary layer inherits the accident layer's method but not its regulatory category; and a result carries forward only where it does not rely on A1–A3.

The layer in effect is an input to AI governance, not a rule of it: it changes whether a Broker can be reached, which mode of discharge carries the load, and which inputs can be trusted. Sections: function and safety §4.2 · accident layer §4.3 · adversary §4.4 · success §4.5 · relations §4.6 · limits §4.7 · cyber §4.9.

FD-DBL_Figure_1_Design_Basis_Layers_v0_1.png

Figure 1 — The design basis layers at a glance

Figure 1 — The design basis layers at a glance. Source file: Graphics/FD-DBL_Figure_1_Design_Basis_Layers_v0_1.svg.

4.7 The operating limits

A design basis is kept by operation. The people, processes, and technologies that run a facility keep it inside the design, and monitoring and surveillance establish that it remains there. They do so against a ladder of limits, each set inside the next.

The rungs. From the inside out, the ladder runs from the normal operating band, through an administrative limit set by the operator with margin to the limit beyond it, to the limiting conditions for operation and the settings of automatic protection, to the limits assumed in the safety analysis, to the safety limits that protect the physical barriers against uncontrolled release. Each rung is a set contained in the next. The distance between rungs is margin, and the administrative limit exists to preserve the margin between ordinary operation and the first limit a regulator enforces. A common practice is to set the administrative limit at a fixed fraction of the limit beyond it, such as eighty percent.

The triple. Each rung is paired with three things: who acts, what the action is, and what is reported. Approaching the administrative limit, the operator or control system acts to stay inside it, and the record is the operator's own, carried in its corrective action program. When a limiting condition for operation is not met, the facility shuts down or follows the remedial action its technical specifications permit until the condition is met again. Automatic protection is set so that it corrects the abnormal condition before a safety limit is exceeded. If a safety limit is exceeded, the facility shuts down and the regulator is notified. A rung without its response and its report is not fully specified, in the same way that a Bright Line without its Broker is not (*The Broker* §4.2).

The documents. The ladder is carried by a chain of documents. The safety analysis report establishes the analyzed design basis, the license rests on it, the technical specifications turn it into operating rules, the procedures carry those rules out, and event reports record departures from them. The chain has the same shape as the authority hierarchy of this corpus, in which definitions are applied by frameworks, frameworks by instruments, and departures are found and recorded by conformance.

The nuclear instance. In the United States, 10 CFR 50.36 requires technical specifications in five categories: safety limits, limiting safety system settings, and limiting control settings; limiting conditions for operation; surveillance requirements; design features; and administrative controls. For a power reactor the technical specifications are Appendix A to the operating license. If a safety limit is exceeded, the reactor must be shut down and the Commission notified. When a limiting condition for operation is not met, the licensee shuts down or follows the remedial action the specifications permit. Under 10 CFR 50.73, a licensee event report is due within sixty days of discovery for specified events, including the completion of a shutdown the technical specifications require, operation or a condition the technical specifications prohibit, a seriously degraded or unanalyzed condition, and the actuation of specified safety systems. Immediate notifications are governed by 10 CFR 50.72. Crossing an administrative limit is not, by itself, reportable to the regulator. Each sector names its own rungs and its own reporting regime; this instrument fixes the structure, not the values.

Placement of the term. "Envelope" is reserved for its authorization sense by FD-EV (*The Envelope*). This instrument uses "operating limits" and "operating band" for the operational sense, and permits a sector's established term, such as the nuclear "design basis envelope," where FD-EV §4.3 permits it.

The ladder across the layers. The ladder belongs to the Design Basis layer, where operation stays in the band. It is exercised at the accident layer, where the response returns operation to within the limits or takes the facility to a safe condition. At the adversary layer it is relied on and weakened at once. The responses and reports at its outer rungs assume that the regulator, the reporting chain, and outside support are present, which is A3, and denial removes them. Those responses must be re-established under the carry-forward rule (§4.6), and the report at those rungs becomes a record kept for later rather than a notice delivered at the time. The MA Trigger Framework is the adversary-layer counterpart of the technical specifications, and should be read as defining the rungs, responses, and records that hold under denial.

Margin under pressure. The gap between the operating band and the administrative limit is where a system rewarded for performance applies pressure. An optimizer that is never permitted to cross a limit can still wear that margin down, pushing operation steadily toward the administrative limit, so that ordinary disturbances begin to reach the limits a regulator enforces. This instrument names the gap as the place where that pressure acts. Whether a given optimizing action is above or below the Bright Line is decided by FD-BL (*The Bright Line*), on that action's consequence, and not here.

4.8 The layer as an input to AI governance

This instrument does not govern the AI Governance line. The rules that govern AI in the corpus are owned elsewhere: placement by FD-BL (*The Bright Line*), the Broker by FD-BR (*The Broker*), the mode of discharge by FD-BL-D1 (*Mode of Discharge*), and indication integrity by FD-BL-D2 (*Indication Integrity*). Those rules are the same at every layer. What changes with the layer is the conditions under which they apply. The layer in effect is an input to AI governance, not a rule of it.

Three conditions change with the layer.

Whether a Broker can be reached. At the Design Basis and accident layers, a Command Broker is available to authorize in the moment. At the adversary layer, denial can sever the Broker from the action, which is the condition the MA Introduction identifies at §4.6 as removing the governance layer at the moment it is most needed.

Which mode of discharge carries the load. As the layer moves outward, advance discharge for pre-proven action classes carries more of the obligation, and in-the-moment authorization less. Under denied conditions the pre-authorized envelope becomes immutable under FD-BL-D1 (*Mode of Discharge*).

Which inputs can be trusted. At the adversary layer, intentionality and precision make the inputs to an AI system targets in their own right, and the requirements of FD-BL-D2 (*Indication Integrity*) on traceable and corroborated inputs carry the weight.

This instrument names these conditions and decides none of them. Whether the AI Governance line has accident and adversary layers of its own, in the sense defined here, is a question for that line.

4.9 Cyber across the layers

Cyber is not a layer. Like people, process, and technology, it runs through every layer, and it enters each one differently.

The accident layer's event set grows with experience (§4.3), and cyber is the clearest record of that growth. The design bases of the 1950s through the 1970s knew fire, flood, earthquake, pipe break, and

the failed pump. Digital instrumentation and control brought software common-cause failure into the set. Malicious cyber activity brought cybersecurity programs into United States nuclear power under 10 CFR 73.54. Artificial intelligence now brings its own error classes. Each addition is the accident layer doing what it was built to do.

At the Design Basis layer, the technology that delivers the function includes digital and AI components. A facility that uses AI carries two design bases, the physical design basis and the AI Governance design basis, which the MA Introduction (§4.6) calls the dual design basis. This instrument describes the facility's design basis; how the layer in effect bears on the AI Governance design basis is stated at §4.8.

At the accident layer, the event set includes non-malicious digital events: software common-cause failure, malfunction of digital instrumentation and control, and AI error, including hallucination and behavioral exceedance, in the error tiers the Tiered Assessment Framework defines. A latent defect shared by every instance of a digital component is a common cause and is analyzed as one under A1, even with no adversary present.

At the adversary layer, cyber is one of the means through which the four characteristics act. Intentionality acts through adaptive intrusion. Persistence acts through pre-positioned or dormant access. Precision acts through targeted manipulation of logic, and through the horizontal scalability of shared components (§4.4). Denial acts through disruption of communications and control. Cyber attack is one of the six threat categories of the MA International Framework (§2.5), and the MA Accident Description (§4.1) requires that the ASSC be achievable and maintainable despite a successful cyber attack, alone or combined with kinetic action.

Below the adversary layer, malicious cyber activity by an actor that does not exhibit all four characteristics, such as a criminal ransomware operation, is outside the adversary layer (§6.4) and is addressed through the facility's cybersecurity program.

Against the operating limits, cyber can corrupt the indications by which operation is measured against its limits (§4.7). Whether an indication can be trusted is governed by FD-BL-D2 (*Indication Integrity*), and the protective functions that act on physical law without any reachable authority are defined by FD-LS (*Local Sovereignty*).

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5. Worked Examples

These examples are illustrative and interchangeable.

A designer builds a device to perform a critical function, operated by people using processes and technology. The designer has a Design Basis. Owners of similar devices compare experience and find that fire, flood, earthquake, storm, equipment failure, operator error, and process flaws have each defeated someone's device. They design to withstand that list, and now have a Design Basis Accident layer, with success meaning that the function recovers or the device reaches a safe, known condition from which it can be restored. Then one owner finds an adversary at work who can disrupt the function with intentionality, persistence, precision, and denial. The owners extend their designs once more. They accept that the function may not survive, and design instead so that it fails in a manner that keeps consequences bounded (§4.2–§4.5).

A sector instrument lists five characteristics of military action, adding “simultaneity” to the canonical four. It does not conform. Simultaneity is the product of intentionality and precision (§4.4), and the instrument should cite this instrument rather than restate the set.

A facility credits an accident-layer analysis showing that a loss of off-site power is recovered when a mobile generator is delivered within eight hours. The analysis depends on A3. It does not carry into the adversary layer, and the coping strategy must be re-established on local resources alone (§4.6).

A design for the adversary layer buries and conceals a backup cooling system. The design improves margin against bombardment and surveillance, but it may lengthen the time needed to reach the system for repair after an ordinary accident-layer failure. The claim that adversary-layer features strengthen the facility against every accident-layer event must be shown for the specific design; it is not true by definition (§6.3).

A plant operates with a coolant temperature limit in its technical specifications and an administrative limit set at eighty percent of that limit. A generative optimizer, rewarded for output, recommends setpoint changes that keep operation just inside the administrative limit for weeks. No limit is crossed and nothing is reportable. The margin the administrative limit exists to protect is nonetheless being consumed, and an ordinary transient now reaches the limiting condition for operation that would previously have stayed inside the band (§4.7).

A facility reaches safe shutdown after a loss of off-site power and plans its restart around grid restoration and fuel delivery. That satisfies the accident-layer criterion. Under military action with denial, the same shutdown state, held on stored fuel with no delivery assumed and no restoration in view, must be shown to remain safe for the design-basis duration on local resources alone (§4.5).

Four redundant protection trains run the same firmware on the same digital platform. At the accident layer, a latent defect in that firmware is a software common-cause failure, analyzed as such and answered by diversity (§4.3, §4.9). At the adversary layer, the same shared firmware is horizontally scalable: one exploit defeats all four trains, and every other facility that runs the same version. Diversity that a single action can defeat does not count as diversity at this layer (§4.4).

A well-resourced criminal group shows persistence, precision, and intentionality against a water utility but cannot deny site access or external support. The group does not exhibit all four characteristics, so it is not within the adversary layer (§4.4, §6.4). The threat assessment records the characteristics it does exhibit, and the facility addresses the group through its other security and threat instruments, including its cybersecurity program where the group acts through cyber means, not through this layer.

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6. Reserved and Determined Items

6.1 Term reservation — resolved here

“Design basis layer” is reserved for the construct defined in §4.1, and the short word “layer” within this instrument is read only in that sense. Instruments that cite this one shall use the full term wherever “layer” could be read in the architectural sense of FD-LD (*Layer Decomposition and Determinism*) or as a level of defense in depth. “The four characteristics,” and the terms intentionality, persistence, precision, and denial when used of an adversary, are reserved for the set defined in §4.4. Variant enumerations in existing instruments are non-canonical and shall be conformed by citation at each instrument’s next

revision (§7). The “denied conditions” of FD-BL-D1 (*Mode of Discharge*) are conditions produced by denial in this sense, and that usage conforms. Conformance is staged per instrument revision, following the reservation model used for “Bright Line,” “envelope” (FD-EV), and “Broker” (FD-BR).

6.2 The adversary envelope — carried by the threat assessment

The accident layer’s event set is bounded by experience. The adversary layer’s conditions are bounded by choice: each facility’s threat assessment characterizes the level of each of the four characteristics against which the facility is designed, as the Design Basis Threat does for physical protection. This instrument requires that the envelope be bounded and that the bound be stated. It does not set the bound.

6.3 Dominance — to be shown, not assumed

The MA International Framework at §8.2 and the MA Accident Description at §1.3 state that design features meeting the adversary layer add margin against every accident-layer event. That holds where the adversary-layer design dominates the accident-layer design in every respect relevant to each event. Where an adversary-layer feature works against an accident-layer response — concealment or burial against repair access, independence from external systems (MA Accident Description §4.1) against an external support path the accident analysis credits — the claim must be shown for that facility class. This instrument records the obligation; the facility-class instruments discharge it.

6.4 Partial-characteristic actors — determined

The adversary layer applies only to an actor that exhibits all four characteristics at the level characterized in the facility’s threat assessment (§4.4). An actor that exhibits some but not all of them is not within this layer. How such an actor is addressed — by the facility’s design basis threat, its security program, or another instrument — is outside this instrument. The threat assessment records the characteristics present, so that a change in the actor’s capability that completes the set is recognized when it occurs.

The same rule decides how a real event is recorded. An event in which a state force acted with intentionality, persistence and precision, but without denial, does not reach the adversary layer. A reference event records it as military action that realizes the scenario it realizes, and does not give it the military-action tier, because that tier is the adversary layer and the layer requires denial. The author determined this on 10 October 2026, on the attacks on the Kerch Strait Bridge, and it holds in every series.

A characteristic the record cannot establish is not present for this purpose. An event in which one or more of the four characteristics is not determined on the sources does not reach the adversary layer, however firmly the others stand. A reference event records it as military action that realizes the scenario it realizes, names each characteristic that is present and each that is not determined, and does not give it the military-action tier. If later evidence establishes a missing characteristic, the instrument is revised and the tier is given then. The author determined this on 13 October 2026, on the destruction of the Kakhovka Dam and the strike on the Okhmatdyt National Children’s Hospital, and it holds in every series. The standard is strict because the tier moves the duty to defend from the owner to the state (§4.5): a looser one would let that duty move on evidence the record does not hold.

A state’s surveillance and intelligence operations, and its covert theft operations, including the sale of drugs and the laundering of money, are military action for this purpose. The adversary layer is the domain of a state’s adversarial actions, and an operation a state conducts against another’s interests

belongs to it whether or not it is conducted in war. Such an operation is recorded as military action that realizes the scenario it realizes, and it carries the military-action tier only when all four characteristics are established on the record, as the paragraphs above require. The author determined this on 14 October 2026, on the compromise of the SolarWinds Orion platform (DBA-IT-RE-1) and the theft from Bangladesh Bank's account at the Federal Reserve Bank of New York (DBA-FIN-RE-1), each of which shows intentionality, persistence and precision without denial, and it holds in every series.

The elements at which precision aims (§4.4) need not belong to the facility under analysis. Where a campaign is directed at the elements on which a facility depends, such as its power, its communications or its supply, the four characteristics may be read against that campaign rather than against the facility as an aim. The facility's instrument then records that the facility itself was not shown to be an aim, and does not infer that it was. The author determined this on 14 October 2026, on Ukraine's payment system through the war (DBA-FIN-RE-3), and it confirms the reading already made in Ukraine's air-raid alerting (DBA-ESS-RE-9). It holds in every series.

Denial is the removal of the external support on which mitigation and recovery depend (A3): access to the site, repair, resupply, external emergency response, and communications (§4.4). An actor who falsifies the indications by which operators see the process does not thereby exhibit denial. The falsification is precision acting through the facility's logic (§4.9), and whether an indication may be trusted is governed by FD-BL-D2 (*Indication Integrity*). An event in which the operators were denied a true view of the process while that support remained available does not establish denial, and is recorded under the paragraphs above as military action that realizes the scenario it realizes, without the military-action tier. The author determined this on 17 October 2026, on the cyber attack on the Natanz enrichment plant (DBA-NUC-RE-9), and it holds in every series.

The adversary's aim (§4.4) is read against the function and against the requirements on how it is performed, and the function is not redefined to reach an aim. An actor who takes a facility's material to use against a third party defeats a requirement on how the facility performs its function: its safety and, where a sector states it beside safety, the custody of its material. An actor who seizes a facility defeats the function by taking it from those it is performed for. Each is within the aim of §4.4, and neither makes custody or authority part of the function. Whether such an actor is within the adversary layer is still decided by the four characteristics on the record. The author determined this on 17 October 2026, on the placement of a radioactive source in Izmailovsky Park (DBA-NUC-RE-13) and the seizure and occupation of the Zaporizhzhia Nuclear Power Plant (DBA-NUC-RE-1), and it holds in every series.

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7. Conformance

These are the changes other instruments in the corpus owe so that they conform to this definition. Each is made at that instrument's next revision, by citing this instrument rather than restating it.

MA Introduction (v1.9) — conforming in substance. §2.4 states the accident-layer assumptions and §3 states the canonical four characteristics. At next revision, cite FD-DBL as the governing definition for both. The running header and footer of v1.9 still read v1.8, which should be corrected in the same touch.

MA International Framework (L1, v2.0) — §2.3 enumerates simultaneity in place of precision, and §8.2 enumerates three characteristics. At next revision, replace both enumerations with citations to FD-DBL

§4.4. §2.3's statement of the accident-layer assumptions conforms in substance to §4.3 and should cite it.

MA Trigger Framework (L3, v3.0) — conforming in purpose. At next revision, cite FD-DBL §4.7 and state the trigger tiers as the adversary-layer rungs of the operating limits.

MA Accident Description (L2, v3.0) — §3 distinguishes the ASSC from cold shutdown, which assumes a path back to restart; that distinction conforms to §4.5. §1.3 enumerates adaptive threat behavior separately from intentional causation, and §2.1 enumerates three respects. At next revision, replace both with citations to FD-DBL §4.4, and cite FD-DBL §4.5 where the ASSC acceptance criterion is introduced, naming the ASSC as the nuclear protected state.

DBA-EN Series Introduction and sector frameworks carrying a military action extension — cite FD-DBL as the source of the three-layer progression at next revision.

The MA International Framework (v2.0) and the MA Accident Description (v3.0) are issued final versions with reserved DOIs. Their conformance is a next-version change, never an in-place edit. The detailed diagnosis and replacement text for each instrument belong in the conformance schedule and are not reproduced here.

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8. Author Decisions

In developing the several hundred documents of the corpus, the author has had to make many decisions. Some were made with outside reviewers who bring specialized knowledge of a sector, a process or a concept; many were made by the author alone. The determinations recorded below are of the second kind, and comment on them is invited.

8.1 Determined, 3 October 2026

The canonical adversary set is intentionality, persistence, precision, and denial. The change in success criterion belongs in the Foundational Definitions. The definition is a new instrument in the FD Governing Stack, designated **FD-DBL**, Design Basis Layers. It is held for the edition following the October 2026 Edition. The adversary is defined by capability.

8.2 Determined, 4 October 2026

The operating limits travel with the design basis and are incorporated in this instrument (§4.1, §4.7), not placed in a separate companion definition.

The function is the service function, the thing the facility is designed to do. That it must do it safely is a requirement on the function, not the function itself (§4.2, §4.5).

FD-DBL does not govern the AI Governance line. The layer in effect is an input to AI governance, which changes the conditions under which its rules apply (§4.8). Whether the governance line has layers of its own is left to that line.

The adversary layer requires all four characteristics. An actor exhibiting fewer is not within the layer (§4.4, §6.4).

Cyber is not a layer. It runs through every layer, as people, process, and technology do (§4.9). Common-cause failure is acknowledged in A1 at the accident layer, and from the adversary's side a common cause is horizontally scalable (§4.3, §4.4).

8.2A Determined, 10 October 2026

The military-action tier is not met without denial. An event showing intentionality, persistence and precision without denial is recorded as military action that realizes the scenario it realizes, and does not carry the tier (§6.4). Determined 10 October 2026.

8.2B Determined, 13 October 2026

The military-action tier requires all four characteristics to be established on the record. A characteristic that is not determined counts as not present: the event is recorded as military action that realizes the scenario it realizes, with the undetermined characteristics named, and does not carry the tier (§6.4). Determined 13 October 2026.

8.2C Determined, 14 October 2026

A state's surveillance and covert theft operations, including the sale of drugs and the laundering of money, are military action; the military-action tier still requires all four characteristics established on the record (§6.4). The four characteristics may be read against a campaign on the elements a facility depends on, rather than against the facility as an aim (§6.4). Determined 14 October 2026; the rulings reached Project 0 in Project 11's note of 14 October 2026 (o).

8.2D Determined, 15 October 2026

The military-action tier moves the duty to defend the facility from the owner to the state where the jurisdiction's law so places it, and does not move the duty to reach and hold the protected state; no design takes credit for a response by the state that it cannot show will arrive (§4.5, §6.4). Approved 15 October 2026, as drafted in FD-DBL_Queued_Edit_for_the_November_Cycle_The_Tier_Moves_the_Duty_to_Defend_Not_the_Duty_to_Survive v0.1.

8.2E Determined, 17 October 2026

Denial does not include the denial of a true view of the process. A falsified view is precision acting through the facility's logic, and its trust is governed by FD-BL-D2 (*Indication Integrity*) (§6.4). The adversary of §4.4 sets out to defeat the function or a requirement on how it is performed, and so reaches an actor who takes a facility's material to use against others or seizes the facility, without redefining the function (§4.4, §6.4). Determined 17 October 2026, answering the three questions referred by Project 6 in Note 2026-10-17d, from §9.2 of the *Nuclear Sector Framework* (DBA-NUC-SF) v0.5. The framework's provisional readings lapse.

8.3 Outstanding

None. All author decisions raised by this instrument are determined.

9. Foundational References

FD-BL — *The Bright Line*. Eclectic Technologies, 2026.

FD-BR — *The Broker*. Eclectic Technologies, 2026.

FD-EV — *The Envelope*. Eclectic Technologies, 2026.

FD-LS — *Local Sovereignty*. Eclectic Technologies, 2026.

FD-BL-D1 — *Mode of Discharge*. Eclectic Technologies, 2026.

FD-BL-D2 — *Indication Integrity*. Eclectic Technologies, 2026.

FD-LD — *Layer Decomposition and Determinism*. Eclectic Technologies, 2026.

Design Basis Accident — Military Action: An Introduction to the Framework and Document Architecture (cited here as the MA Introduction). Version 1.9. Eclectic Technologies, 2026.

DBA-MA International Framework (L1; cited here as the MA International Framework). Version 2.0. Eclectic Technologies, September 2026.

DBA-MA Formal Accident Description and Design Requirements (L2; cited here as the MA Accident Description). Version 3.0. Eclectic Technologies, September 2026.

DBA-MA Trigger Framework (L3; cited here as the MA Trigger Framework). Version 3.0. Eclectic Technologies, September 2026.

DBA-MA Monitoring Guidance (L4; cited here as the MA Monitoring Guidance). Version 3.0. Eclectic Technologies, September 2026.

Design Basis for AI Governance: Series Overview. Version 0.1. Eclectic Technologies, 2026.

United States Code of Federal Regulations, Title 10, §50.13 (Attacks and destructive acts by enemies of the United States; and defense activities), §50.36 (Technical specifications), §50.72 (Immediate notification requirements for operating nuclear power reactors), §50.73 (Licensee event report system), and §73.54 (Protection of digital computer and communication systems and networks).

Tiered Assessment Framework. Eclectic Technologies, 2026.

International Atomic Energy Agency. *Fundamental Safety Principles*. IAEA Safety Standards Series No. SF-1. Vienna, 2006.

Revision History

0.16 — 17 October 2026. Answers the three questions Project 6 referred from §9.2 of the *Nuclear Sector Framework* (DBA-NUC-SF) v0.5 in Note 2026-10-17d, on the author's determinations of 17 October 2026, recorded at §8.2E. §4.4 restates the adversary's aim: from an actor who sets out to stop the facility from performing its function, to one who sets out to defeat the function or a requirement on how it is performed, of which safety is the one every facility carries, whether by stopping it, seizing the facility, or turning its plant or material against others, without redefining the function. The wording leaves room for a requirement a sector states beside safety, as the *Nuclear Sector Framework* §1.6 states custody of material. It changes a definition, which the rulings of v0.12 to v0.15 did not. §6.4 gains

two paragraphs. The first declines to extend denial to the denial of a true view of the process, made on Natanz (DBA-NUC-RE-9). The second records the restated aim, made on Izmailovsky Park (DBA-NUC-RE-13) and Zaporizhzhia (DBA-NUC-RE-1). The provisional readings the framework stated on each question lapse. Brought into the November 2026 Edition before deposit, by the author's decision of 17 October 2026, in place of v0.15; recorded in FD-REG v0.33 in the same act. No section renumbered.

0.15 — 15 October 2026. Takes in the queued edit FD-DBL_Queued_Edit_for_the_November_Cycle_The_Tier_Moves_the_Duty_to_Defend_Not_the_Duty_to_Survive v0.1, approved by the author as drafted on 15 October 2026 and recorded at §8.2D. §4.5 gains a paragraph after the adversary-layer success criterion: where the jurisdiction's law places defense against a state's armed forces with the state, as 10 CFR 50.13 does for United States nuclear facilities, reaching the military-action tier moves the duty to defend to the state but not the duty to reach and hold the protected state, because the state's defense is external support (A3) that denial removes; the handoff is an allocation of responsibility and never a credit in the design. §6.4 gains a sentence giving the reason the all-four standard is strict. §9 adds 10 CFR 50.13. The edit's base, v0.13, is read as v0.14 (FD-REG v0.27). No definition changes; no section is renumbered. Issued in the November 2026 Edition, and recorded in FD-REG v0.32 in the same act. This is the instrument's first issue.

0.14 — 14 October 2026. Adds to §6.4 two rulings of the author of 14 October 2026. The first is that a state's surveillance and covert theft operations, including the sale of drugs and the laundering of money, are military action, the tier still requiring all four characteristics on the record; made on the SolarWinds Orion compromise (DBA-IT-RE-1) and the Bangladesh Bank theft (DBA-FIN-RE-1). The second is that the elements precision aims at need not belong to the facility under analysis, so that the characteristics may be read against a campaign on a facility's dependencies; made on Ukraine's payment system through the war (DBA-FIN-RE-3), confirming the reading of Ukraine's air-raid alerting (DBA-ESS-RE-9). Both reached Project 0 in Project 11's note of 14 October 2026 (o). Records them at §8.2C under their date. No change to the definitions, and no section renumbered.

0.13 — 13 October 2026. Adds to §6.4 the author's ruling of 13 October 2026 that a characteristic not determined on the record counts as not present: an event in which one or more of the four characteristics is not determined is recorded as military action that realizes the scenario it realizes, with the undetermined characteristics named, and does not carry the military-action tier until later evidence establishes the set. The question arose on two reference events written by Project 11, The Destruction of the Kakhovka Dam (DBA-DAMS-RE-1), which finds precision not determined, and The Strike on the Okhmatdyt National Children's Hospital (DBA-HPH-RE-3), which finds intentionality and precision not determined, and each of which had given the tier. Records the ruling at §8.2B under its own date. No change to the definitions, and no section renumbered.

0.12 — 13 October 2026. Adds to §6.4 the author's ruling of 10 October 2026 that the military-action tier is not met without denial: an event in which a state force acted with intentionality, persistence and precision, but without denial, is recorded as military action that realizes the scenario it realizes, and does not carry the tier. The ruling was made on the attacks on the Kerch Strait Bridge (DBA-TRAN-RE-11, since recast as an illustration in Ways and Linehaul) and reached Project 0 in Project 9's note of 10 October 2026, which asked that it be recorded where the tier is stated. It applies the all-four rule this instrument already determined (§6.4, §8.2) to the recording of real events, and records the ruling at §8.2 under its own date. No change to the definitions, and no section renumbered.

0.11 — 4 October 2026. Takes in the author's edits of 4 October: the full name of the instrument at §2.2, the assumptions placed in parentheses at §2.3, protected states named by facility classes as well as

sectors at §2.3, and “cybersecurity” as one word throughout the body. Adds opening paragraphs to §7, stating what the conformance list is and how it is discharged, and to §8, stating whose decisions are recorded there and inviting comment. The worked examples at §5 remain paragraphs. No change to the definitions.

0.10 — 4 October 2026. Adds Figure 1, The design basis layers at a glance, at the end of §4.6, with a paragraph stating what it shows and that the text governs where the two differ. The figure is held as its own file in Project 0’s Graphics folder. No substantive content changes.

0.9 — 4 October 2026. Revises the citation form of v0.8. A section citation now gives the title and section alone, as in *The Broker* §4.2, without the designation in front; where the running text names an instrument, it gives the designation with the title in parentheses, as in FD-BR (*The Broker*). The DBA-MA series is cited by short title — the MA Introduction, MA International Framework, MA Accident Description, MA Trigger Framework and MA Monitoring Guidance — with full titles and the L1–L4 designations in §9. Restates the citation-form note at §2.3 accordingly. No substantive content changes.

0.8 — 4 October 2026. Gives every citation to another instrument its title alongside its designation, so that a section number reads as a section within a named document, and states the citation form at §2.3 with the L1–L4 designations expanded. No substantive content changes.

0.7 — 4 October 2026. Expands §2.3 into a declaration of every construct the instrument introduces — the design basis layer, the function, safety, the assumptions, the adversary, the characteristics, the success criterion, the protected state, the operating limits, and cyber — each with the section that defines it, so that the instrument’s ontology reads in one place. Adds to the declaration of a design basis layer that it is not an architectural layer in the sense of FD-LD, not a boundary between architectural layers, and not a level of defense in depth. Extends the §6.1 reservation so that the short word “layer” within this instrument is read only as a design basis layer, and requires citing instruments to use the full term wherever the architectural or defense-in-depth sense could be read.

0.6 — 4 October 2026. Adds §4.9, Cyber across the layers, on the author’s determination that cyber runs through every layer rather than forming one: the growth of the accident layer’s event set from physical events through software common-cause failure, malicious cyber and AI error; the dual design basis at the Design Basis layer; non-malicious digital events at the accident layer; cyber as a means of the four characteristics at the adversary layer; malicious cyber below the adversary layer as a matter for the cyber security program; and the corruption of indications against the operating limits. Refines A1 to acknowledge common-cause failure as analyzed at the accident layer (§4.3), and adds to precision that a common cause is horizontally scalable from the adversary’s side (§4.4). Adds a shared-firmware worked example, the cyber security program to the partial-actor example, the determination at §8.2, and 10 CFR 73.54 and the Tiered Assessment Framework to the references.

0.5 — 4 October 2026. Confirms the all-four requirement: the adversary layer applies only to an actor exhibiting all four characteristics, and an actor exhibiting fewer is outside the layer and addressed through the facility’s other security and threat instruments (§6.4, retitled from reserved to determined). Revises the partial-capability worked example to match. Records the determination at §8.2; no author decisions remain outstanding.

0.4 — 4 October 2026. Adds §4.8, The layer as an input to AI governance: FD-DBL does not govern the AI Governance line; the governance rules owned by FD-BL, FD-BR, FD-BL-D1 and FD-BL-D2 are the same at every layer, and the layer changes the conditions under which they apply, namely whether a Broker can be reached, which mode of discharge carries the load, and which inputs can be trusted. Records the

determination at §8.2 and removes the governance-line item from the outstanding decisions; whether the governance line has layers of its own is left to that line.

0.3 — 4 October 2026. Records the author's determination that the function is the service function, the thing the facility is built to do, and that safety is a requirement on how the function is performed rather than the function itself (§4.2, §8.2). Adds to §4.5 that the facility may give up its function at the adversary layer but never the safety requirement, which is the one obligation that survives every layer unchanged. Removes the "what function means" item from the outstanding decisions.

0.2 — 4 October 2026. Corrects the accident-layer success criterion: the response either recovers operation to within the limits or reaches a safe, known condition, which at that layer is a waypoint reached with external support (§4.3). Restates the change at the adversary layer accordingly: recovery drops out as a success outcome and the safe condition must be self-sustaining and indefinite (§4.5). Adds the operating limits as the fifth part of a layer (§4.1) and defines them at §4.7: the ladder of nested limits from operating band to safety limit, the who-acts, what-action, what-report triple at each rung, the document chain from safety analysis report to event report, the United States nuclear instance under 10 CFR 50.36, 50.72 and 50.73, the use of "operating limits" in place of "envelope" for the operational sense, the ladder's behavior across the three layers with L3 as the adversary-layer counterpart of the technical specifications, and the administrative margin as the place where optimization pressure acts. Adds two worked examples, L3 to conformance, and the 4 October determination.

0.1 — October 2026. Initial draft. Defines a design basis layer as function, conditions, assumptions, and success criterion (§4.1). Establishes the function as the invariant of the progression and the Design Basis as the only defining layer (§4.2). States the accident layer's three assumptions — independence, bounded progression, available support (§4.3). Fixes intentionality, persistence, precision, and denial as the canonical adversary characteristics, each defined as acting on the accident set and each removing a stated assumption, with the adversary defined by capability (§4.4). Defines the change in success criterion from delivery of the function to bounded consequence in a protected state, with the ASSC as the nuclear instance (§4.5). States the four relations carried by the layer arrow and the carry-forward rule (§4.6). Reserves the layer terms and the four characteristics, assigns the envelope bound to the threat assessment, records the dominance obligation, and reserves the partial-characteristic case (§6).